

TECHNICAL DOCUMENTATION

Service: k8s_mailfrom

mailfrom-milter

Postfix milter written in Go that enforces domain alignment between the SMTP envelope sender (MAIL FROM) and the From: message header for authenticated sessions. Prevents DKIM domain spoofing: an authenticated user setting MAIL FROM to their own domain but forging From: with a victim domain would produce a valid DKIM signature for a domain they do not own. This milter rejects such messages before DKIM signing occurs.

Only authenticated SMTP sessions (SASL) are checked. Unauthenticated connections (inbound MX delivery) pass through without inspection.

Architecture

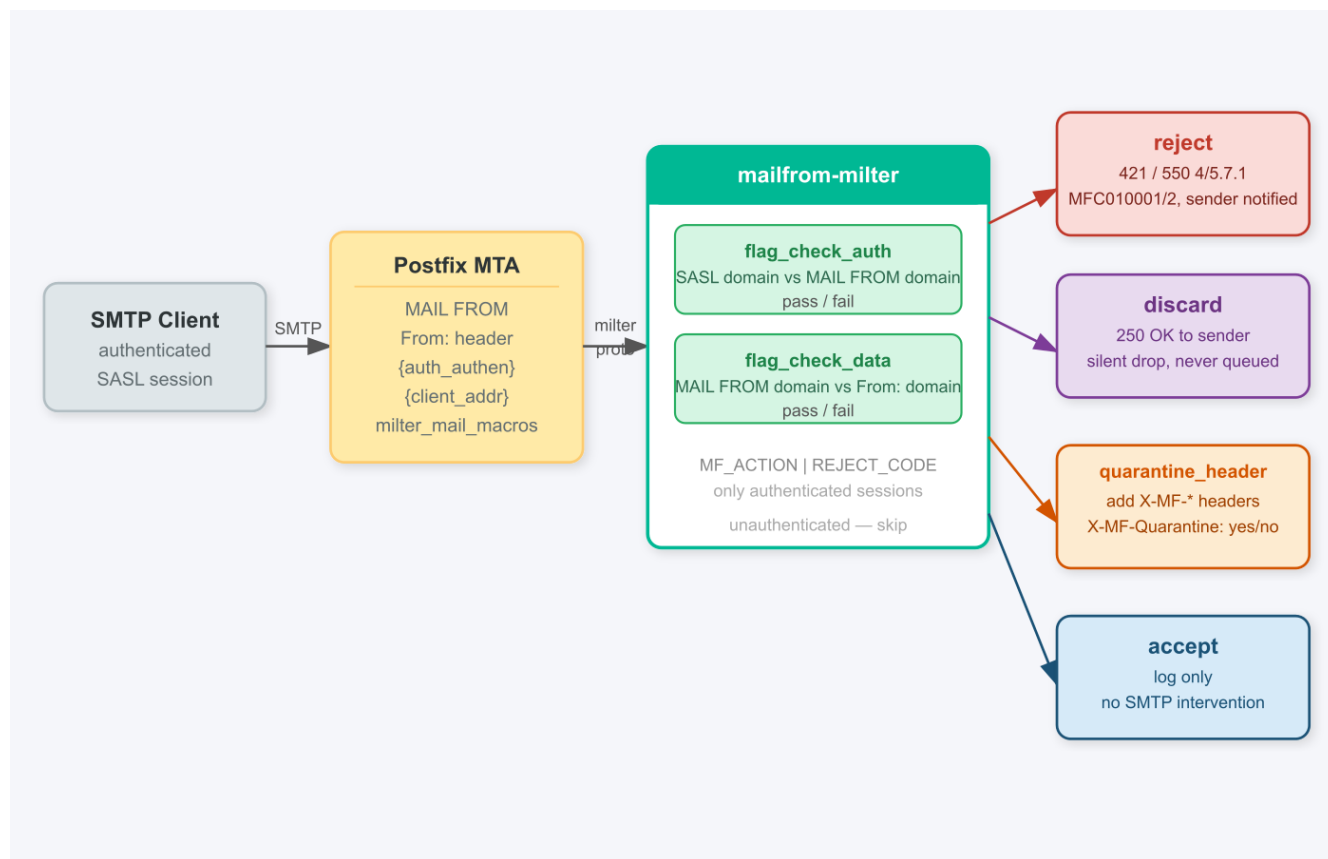


Figure 1: mailfrom-milter architecture

Stack

- Go 1.26 (alpine:3.21 runtime image)
- github.com/emersion/go-milter v0.4.1
- log/slog — structured JSON logging

Directory layout

k8s_mailfrom/

```

|- app/go/
|   |- main.go
|   |- Dockerfile
|   |- go.mod
|   \- go.sum
|- helm/
|   |- Chart.yaml
|   |- values.yaml
|   \- templates/
|       |- deployment.yaml
|       \- service.yaml
|- helm_values/
|   \- values-micro-seconddns.yaml
|- doc/
|- argocd-app.yaml
\-.github/workflows/ci.yaml

```

Checks

For every authenticated session two checks are performed:

Flag	Check	Values
flag_check_auth	SASL username domain vs MAIL FROM domain	pass / fail
flag_check_data	MAIL FROM domain vs From: header domain	pass / fail

Actions

Configured via the MF_ACTION environment variable:

Action	On check fail	On both pass
reject	SMTP reject with REJECT_CODE 4/5.7.1	log + accept
discard	silent drop (250 OK to sender), log	log + accept
quarantine_header	add X-MF-Quarantine: yes headers, log	add X-MF-Quarantine: no headers, log
accept	log only, accept	log only, accept

Headers added for quarantine_header action:

Header	Value
X-MF-Envelope-From	MAIL FROM address
X-MF-From	Address extracted from From: header
X-MF-Quarantine	yes if any check failed, no if all passed

Environment Variables

Variable	Default	Description
LISTEN_ADDR	0.0.0.0:10031	TCP address to listen on
MF_ACTION	reject	reject / discard / quarantine_header / accept
REJECT_CODE	421	SMTP reply code for reject: 421 (temp) or 550 (permanent)
LOG_LEVEL	—	Set to debug for verbose per-message logging

Deploy

Managed by ArgoCD. Apply the ArgoCD Application manifest:

```
kubectl apply -f argocd-app.yaml
```

Postfix configuration:

```
smtpd_milters = inet:mailfrom.mail.svc.cluster.local:10031
milter_mail_macros = i {mail_addr} {client_addr} {client_name} {auth_authen} {auth_type}
milter_default_action = accept
```

Local development:

```
docker build -t mailfrom:dev app/go/
docker run --rm -p 10031:10031 -e MF_ACTION=accept -e LOG_LEVEL=debug mailfrom:dev
```

Logging

Every processed authenticated message produces one JSON log entry:

```
{
  "time": "...", "level": "INFO", "msg": "milter",
  "envelope_from": "user@attacker.com",
  "auth_user": "user@attacker.com",
  "flag_check_auth": "pass",
  "from_header": "ceo@victim.com",
  "flag_check_data": "fail",
  "return_code": "reject"
}
```

return_code values: reject, discard, accept. Enable verbose logging without code changes: LOG_LEVEL=debug.